Speech-Eval: technical appendix

Companion to the final report (docs/report/report.pdf). Sriram MPS, IIT Mandi. Full derivations, detailed tables, the diagnosis, the pause-floor check, the next-step mathematics and every constant. Section numbers here are letters; figure numbers start with T.

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A. The mathematics, step by step

This section gives every formula the detector uses and explains how we arrived at it. Everything matches the frozen code (`features.py`, `reference.py`, `detect.py`, `score.py`, `evaluate.py`). Section F lists every constant.

A.1 What we measure

Pitch $f_0(t)$ and intensity $I(t)$ are computed with Praat (two-pass autocorrelation pitch) on a 10 ms grid. Pitch is converted to **semitones** relative to the speaker's own median pitch:

$$\mathrm{st}(t) = 12 \log_2 \frac{f_0(t)}{\tilde{f}_0}$$

Twelve semitones is one octave, so a deep voice and a high voice that move by the same musical interval get the same numbers. Each word window w (the word and one neighbour on each side) then gets four measurements:

$$\mathrm{rate}_w = \frac{\sum_{j \in w} \mathrm{syl}_j}{(t_w^{\mathrm{end}} - t_w^{\mathrm{start}}) - \sum_{g \in w,\ g > 0.25\,\mathrm{s}} g}$$

$$\mathrm{pitchvar}_w = \mathrm{SD}\{\mathrm{st}(t) : t \in w\}$$

$$\mathrm{loud}_w = \mathrm{mean}_{t \in w} \left(I(t) - \tilde{I}\right) \ \ \mathrm{dB}$$

$$\mathrm{clarity}_w = \mathrm{mean}_{t \in w}\ 10 \log_{10} \frac{P_{2\text{-}6\,\mathrm{kHz}}(t)}{P_{0\text{-}2\,\mathrm{kHz}}(t)} \ \ \mathrm{dB}$$

and each boundary j gets a pause:

$$\mathrm{pause}_j = t_j^{\mathrm{start}} - t_{j-1}^{\mathrm{end}}$$

Syllables come from vowel phones in the CMU pronouncing dictionary. Gaps longer than 0.25 s are removed from the rate denominator, so a long pause is not counted as slow speech (pauses have their own detector). Loudness is relative to the speaker's own median loudness, which cancels microphone gain. Clarity uses the fact that muffled speech loses energy above 2 kHz, where consonants such as s, t and f live.

A.2 Why logarithms

Speaking 20% faster means the same thing for a slow reader and a fast reader. On a log scale a 20% change is always the same distance:

$$\ln(1.2\,r) - \ln r = \ln 1.2 \quad \text{for every } r$$

so one spread can describe readers of very different natural speeds. Logs also turn right-skewed, positive quantities (rates, spreads, pause lengths) into roughly symmetric ones, which suits a z-score. The transforms are:

$$T_{\mathrm{rate}}(x) = \ln x, \quad T_{\mathrm{pitchvar}}(x) = \ln(x + 0.25), \quad T_{\mathrm{pause}}(x) = \ln(x + 0.05), \quad T_{\mathrm{loud}} = T_{\mathrm{clarity}} = x$$

Pauses and pitch variation can be exactly zero, and ln 0 is undefined. Adding a small offset (0.05 s; 0.25 semitones) keeps zero finite and squeezes together very small values that lie within measurement noise. Loudness and clarity are already in decibels, which is a log scale, so they are used as they are.

A.3 A robust centre and spread

For each word and feature, the reference readers give values $x_1 \ldots x_n$ (after the transform). A mean and standard deviation would let one unusual reader (say, one who stopped to breathe) drag the reference. We use the median and the median absolute deviation (MAD), which ignore up to half of the readers being unusual:

$$\mathrm{MAD} = \mathrm{median}_i \, |x_i - \mu|, \qquad \mu = \mathrm{median}_i \, x_i$$

To put the MAD on the same scale as a standard deviation, we ask what it would be for normally distributed data. Half of a normal distribution lies within 0.6745 σ of the mean, so:

$$X \sim \mathcal{N}(\mu, \sigma^2) \;\Rightarrow\; \mathrm{median}\,|X - \mu| = \Phi^{-1}(0.75)\,\sigma = 0.6745\,\sigma \;\Rightarrow\; s = \frac{\mathrm{MAD}}{0.6745} = 1.4826\,\mathrm{MAD}$$

This is where the constant 1.4826 in the code comes from.

A.4 Few readers: borrowing strength (shrinkage)

Mode A for Gettysburg has 10 readers; Reagan and all of Mode B have one. With one reader, s = 0 and a z-score would divide by zero. Even with ten readers, a word where all of them happen to agree gives s near 0. We fix this by combining the word's own spread with a **pooled spread** learned from all words, treating the pooled value as if it came from $n_0 = 5$ extra imaginary readers. Pooling the variances of two groups (n real readers with variance s^2, n_0 imaginary readers with variance σ_{pool}^2) gives:

$$\sigma^2 = \frac{n\,s^2 + n_0\,\sigma_{\mathrm{pool}}^2}{n + n_0}, \qquad n_0 = 5$$

This is the standard pooled-variance formula, and it can also be read as a Bayesian prior on the variance worth five observations. With many readers the word's own spread dominates; with few, the pooled spread does. The pooled spread for each feature f is the median, over all words, of the per-word spreads of the 18 training readers, with a numerical safeguard of 0.001:

$$\sigma_{\mathrm{pool},\,f} = \max\left(0.001,\ \mathrm{median}_{\mathrm{words}}\ s_{w,\,f}\right)$$

Feature	rate	pitchvar	loud	clarity	pause
σ_{pool} (transformed units)	0.1634	0.3071	2.859 dB	4.726 dB	**0.001** (safeguard)

Two special cases matter later:

$$n = 1,\ s = 0: \quad \sigma = \sqrt{5/6}\ \sigma_{\mathrm{pool}} = 0.913\,\sigma_{\mathrm{pool}}$$

$$n = 10,\ s = 0: \quad \sigma = \sqrt{5/15}\ \sigma_{\mathrm{pool}} = 0.577\,\sigma_{\mathrm{pool}}$$

Keep this in mind

For pauses, the median per-word spread is exactly zero, because at most word boundaries no reader pauses at all. So the pooled pause spread is only the safeguard value 0.001, and shrinkage pulls individual boundaries even lower (0.000577 to 0.000913) wherever no reader pauses. Section C shows that this explains the flood of awkward-pause flags.

A.5 The z-score and the threshold k = 3.5

$$z = \frac{T(x) - \mu}{\sigma} \qquad \text{(rounded to 4 significant digits)}$$

Each flaw type looks at one feature in one direction (d = +1 for rushing, −1 for the others):

$$\text{word flagged for type } \tau \Leftrightarrow d_\tau \, z_{f(\tau)} \geq k_\tau, \qquad d_\tau \in \{+1, -1\}, \; k_\tau = 3.5$$

How k was chosen. On the development speakers only, each type's k was swept from 1.5 to 3.5 in steps of 0.25, one type at a time, for up to three rounds, keeping the value with the best F1; ties went to the more conservative value. All seven types chose 3.5, which is the top of the grid we fixed in advance. The pre-declared guard of at most 0.5 false alarms per minute on clean development recordings was **not** met (1.777 per minute). We report this as a failed acceptance check.

What 3.5 should mean. If the reference spread were correct and values were roughly normal, a clean word would pass 3.5 by chance very rarely:

$$P(Z \geq 3.5) = 1 - \Phi(3.5) \approx 2.3 \times 10^{-4} \quad (\text{vs. } 2.3 \times 10^{-2} \text{ at } k = 2)$$

A minute of reading has about 150 words, so roughly 900 one-sided word-feature tests (rate is tested in both directions): 900 × 2.3×10^{-4} ≈ 0.21 chance flags per minute, before the three-word rule lowers it further. The observed rate (1.3–1.8 per minute) is much higher. That gap is evidence that some spreads are far too small, which is exactly what Section C finds for pauses.

A.6 From words to regions

For the five non-pause types, flagged words are grouped into runs inside one sentence. A run may skip at most one unflagged word; an unmatched or invalid word ends it. Runs of at least three flagged words become a region, from the start of its first flagged word to the end of its last. The region's evidence is the mean of the flagged words' transformed values, converted back to physical units (syllables per second, semitones, dB) for the explanation. Why three words? A single window can spike from one mis-measured frame, and neighbouring windows share words, so one bad word can affect three windows; requiring three flagged words in a run reduces such isolated spikes.

A.7 Pause events

A pause is compared only when the two words around it are also neighbours in the reference. Each rule has a statistical part (the z-score) and a physical part, so that a flag is both surprising *and* large enough to hear:

- **awkward_pause:** z ≥ 3.5 **and** the pause lasts ≥ 0.45 s.
- **missing_pause:** z ≤ −3.5 **and** the reference pause is ≥ 0.3 s **and** yours is ≤ 40% of it.
- Gaps shorter than 30 ms are drawn as a 30 ms event centred on the boundary, so they can be seen and matched.

- Three containment rules remove double counting: a missing pause strictly inside a rushing region, an awkward pause strictly inside a dragging region, and a too-quiet region inside a mumbling region with a smaller mean |z|. Pause flags do not otherwise suppress any other type; every type is detected independently.

A.8 Severity

$$\text{severity} = \text{clip}\Big(\#\big\{b \in B : b \le \overline{|z|}\big\},\ 1,\ 5\Big), \quad B = (2.0,\ 2.75,\ 3.5,\ 4.059,\ 6591)$$

Here |z|-bar is the mean |z| of the region's flagged words. The cut points were set on development data. Because every flagged word already has |z| ≥ 3.5, Mode A and B regions in practice receive severity 3, 4 or 5. On development data the rank correlation between predicted and injected severity was weak (Spearman 0.157), so severity is a rough ordering, not a calibrated scale.

A.9 The score card

Each category starts at 10. Duration flaws (pace, pitch, volume, clarity) cost points in proportion to how much of the speech they cover; pauses cost a fixed amount per event. D is the speech duration from the first to the last word.

$$\text{penalty}_c = \sum_{r \in c} \text{sev}_r \cdot \frac{\text{end}_r - \text{start}_r}{D} \cdot 10$$

$$\text{penalty}_{\text{pauses}} = \sum_{r} 0.4 \cdot \text{sev}_r$$

$$\text{score}_c = \text{round}_{0.1}(\text{clamp}(10 - \text{penalty}_c,\ 0,\ 10)), \qquad \text{overall} = \text{round}_{0.1}\left(\sum_c w_c\, \text{score}_c\right)$$

Category	Pace	Pitch	Pauses	Volume	Clarity
Flaw types	rushing, dragging	monotone	awkward, missing pause	too_quiet	mumbling
Weight w_c	0.25	0.25	0.20	0.15	0.15

So a severity-s region covering a fraction φ of the speech costs 10·s·φ points. For example, severity-3 rushing over 10% of the speech gives

$$\text{pace} = 10 - 3 \times 0.10 \times 10 = 7.0$$

and a severity-3 pause costs 1.2 points, so nine such pauses empty the pause category. For the dashboard example in Figure 3 of the main report (13 pause regions, nothing else):

$$\text{overall} = 0.25(10) + 0.25(10) + 0.20(0) + 0.15(10) + 0.15(10) = 8.0$$

which matches the 8.0 shown on screen. Rounding uses exact decimal arithmetic with half-up rounding, so the same input always gives the same score. The weights are design choices (pace and pitch weighted most), not a validated perceptual scale.

A.10 How we measure the detector

A predicted region counts as correct (a true positive) if it has the same type as an injected flaw and overlaps it enough. Overlap is measured by intersection over union:

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Pairs are matched greedily, highest IoU first, each prediction and each label used at most once, with IoU ≥ 0.3. For injected pause events, which are very short, a pair is eligible when the centres are within 0.25 s. On the author's takes we used the stricter rule (IoU ≥ 0.3 for every type). Then:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad F_1 = \frac{2PR}{P+R}$$

Worked check for A-LOSO: 82 hits, 123 misses and 576 extra flags give P = 82/658 = 0.125, R = 82/205 = 0.400 and F_1 = 0.190, as reported. A second measure of false alarms uses only recordings with no edits:

$$\text{clean alarms/min} = \frac{\#\text{ flags on unedited recordings}}{\text{minutes of unedited speech}}$$

Confidence intervals. We resample the test *files* with replacement 1,000 times (seed 20261001), recompute the pooled metric each time, and take the 2.5th and 97.5th percentiles:

$$\mathrm{CI}_{95} = \left[\, q_{0.025}(\hat{\theta}^*_1 \ldots \hat{\theta}^*_{1000}),\ q_{0.975}(\hat{\theta}^*_1 \ldots \hat{\theta}^*_{1000}) \,\right]$$

Edited files from the same speaker are correlated, so these are file-level intervals, not speaker-level ones. **Monotonicity** of scores is checked with Spearman's rank correlation between injected severity and the category score; it should be negative (worse flaw, lower score).

B. Detailed benchmark results

All numbers in this section are for the frozen v1 detector on the **test** speakers, evaluated once at commit aaf1832. Thresholds were chosen on development speakers only.

Setting	Files	Precision	Recall [95% CI]	F1 [95% CI]	Clean alarms / min
A-LOSO (expert readers)	190	0.125	0.400 [0.328, 0.471]	0.190 [0.156, 0.223]	1.341
B-same-speaker (own original)	76	0.018	0.390 [0.289, 0.494]	0.035 [0.026, 0.045]	0.000

By flaw type (A-LOSO)

Type	Precision	Recall [95% CI]	F1 [95% CI]	Clean alarms / min
rushing	0.625	0.167 [0.038, 0.312]	0.263 [0.067, 0.450]	0.000
dragging	1.000	0.185 [0.055, 0.343]	0.312 [0.105, 0.511]	0.000
monotone	0.100	0.419 [0.240, 0.606]	0.161 [0.087, 0.242]	0.335
too_quiet	0.714	0.179 [0.040, 0.333]	0.286 [0.069, 0.487]	0.000
mumbling	1.000	0.483 [0.304, 0.667]	**0.651** [0.467, 0.800]	0.000
awkward_pause	0.156	**0.903** [0.778, 1.000]	0.265 [0.197, 0.332]	0.447
missing_pause	0.038	0.414 [0.240, 0.600]	0.070 [0.037, 0.108]	0.559

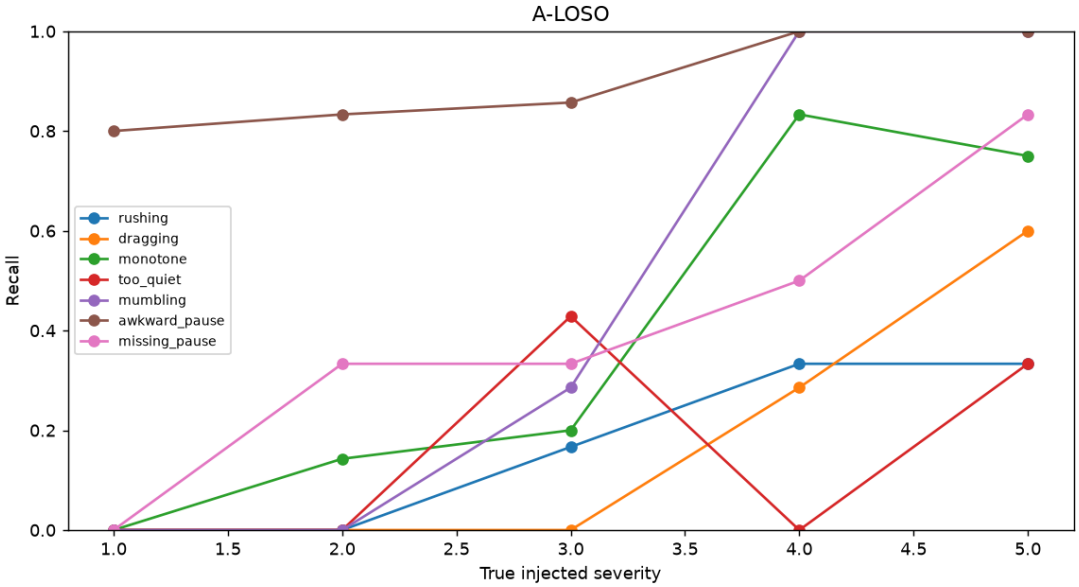


Figure T2. Recall rises with injected severity for most types (A-LOSO, test). Awkward pauses are found even at severity 1; rushing and too_quiet stay low.

Reading the results. Pause flaws are the easiest to find because a pause is measured directly in seconds at one boundary. Non-pause flaws must change a three-word window by more than 3.5 typical deviations of real readers, and then form a run of three flagged words; mild edits often do not. Mumbling has the best balance (F1 0.651, no clean alarms). Most false alarms are pause flags (awkward 0.447 and missing 0.559 per clean minute).

In B-same-speaker, recall is similar (0.390) but precision collapses to 0.018: 1,390 of 1,771 flags are awkward pauses. With a single reference recording, every boundary borrows the tiny pooled pause spread, so almost any timing difference passes 3.5 (Section C). B's clean-alarm rate of 0 should not be read as a strength: there the reference is the speaker's own original, which leaves a clean file little room to differ.

Do the scores fall when flaws get worse?

For every type, the median Spearman correlation between injected severity and the matching category score is negative (−0.71 to −0.89), so stronger flaws give lower scores. Some series are constant (no flag at any severity), and monotone is non-increasing in only 29% of series, so the scores are only partly sensitive.

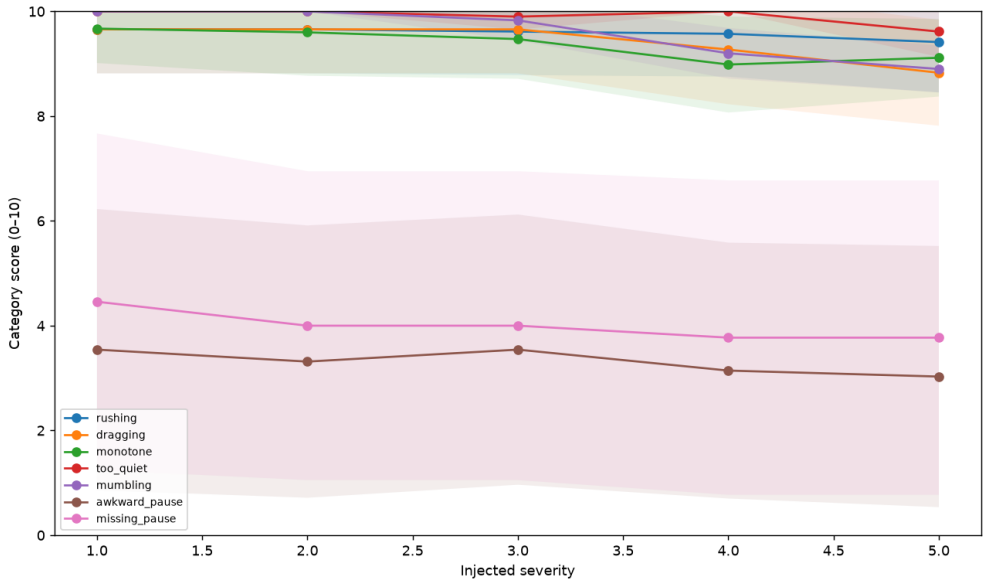


Figure T3. Category score against injected severity.

A post-hoc second detector (v2)

After v1 was published we tried, on development data only, a second detector (v2) that also requires a word to differ from the speaker's own other sentences and to exceed a minimum practical effect. We logged all 30 configurations, then ran the selected one once on the test set. None of the first 29 configurations met the clean-alarm guard; the selected configuration 30 was an explicitly adaptive repair that raised the awkward-pause excess to 1.10 s using development data only, and it is disclosed as such. Because the test set had already been seen with v1, these results are optimistic.

Setting	Version	Precision	Recall	F1	Clean alarms / min
A-LOSO	v1 (primary)	0.125	0.400	0.190	1.341
A-LOSO	v2 (post-hoc)	0.145	0.288	0.193	0.671
B-same-speaker	v1 (primary)	0.018	0.390	0.035	0.000
B-same-speaker	v2 (post-hoc)	0.042	0.207	0.070	0.000

v2 halves clean alarms in Mode A but loses a quarter of the recall, and still fails the 0.5 per minute guard. It is a trade-off, not an improvement, so v1 stays primary.

Mode C against same-text comparison

Mode C has no same-text reference, so it compares sentence-level rate, pitch range, final loudness drop and sentence-end pause with norms from the training speakers (threshold 2.0). On 175 paired test files, scored at sentence level:

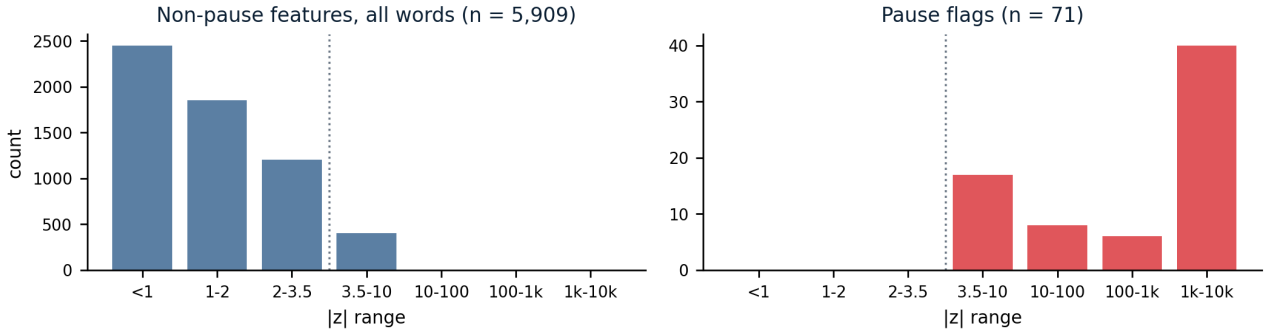
System	Precision	Recall	Clean alarms / min
Same-text reference (A/B)	0.106	0.324	0.584
Mode C (norms only)	0.020	0.179	4.159

Mode C is clearly weaker, which is why the dashboard labels its extra tips “heuristic, not scored”. Training speakers' median sentence rates range from 2.9 to 5.2 syllables per second, so a single norm cannot fit everyone.

C. Diagnosis: why mostly pause flags

We ran a read-only diagnosis of the frozen v1 outputs on the author's takes (no settings changed). Across the six scored takes in Mode A, 71 flags were pauses and 10 were anything else. Pause flags dominate for two different reasons.

- **Missing pauses** (55 flags in Mode A; 53 of the 77 false alarms): at sentence boundaries where the expert readers pause at least 0.3 s, the author paused less than 40% as long. These flags pass both the statistical and the physical rule. Whether a listener would call them flaws is unknown without independent ratings.
- **Awkward pauses** (16 flags in Mode A; most of the flags in Mode B): these come from a near-zero reference spread, explained below. This is the cause that a timing floor can address (Section D).



Mode A, six scored genuine takes: dotted line = flag threshold 3.5

Figure T5. Left: non-pause z-scores for every word are mostly below 3.5. Right: pause flags have z-scores mostly far above 10 (54 of 71; median 1,150).

The chain of cause and effect

- At most word boundaries, no reference reader pauses, so the per-word pause spread is exactly 0.
- The pooled pause spread is the median of those spreads, which is 0, so only the numerical safeguard 0.001 remains (Section A.4).
- At boundaries where no reader pauses, shrinkage pulls σ down to between 0.000577 and 0.000913 in log units. (Boundaries where readers do pause have much larger spreads, up to 1.47.)
- With σ = 0.000913, a 10 ms difference around a 0.5 s pause gives z ≈ 20 (the smallest stored case, 0.49 s against 0.48 s, gives z = 20.5), and near zero it would give about 200. The 0.45 s length rule is then the only thing deciding whether it is flagged.

The real “larger | sense” pause shows the scale. The reference had no pause, the take had 1.57 s:

$$z_B = \frac{\ln(1.57+0.05) - \ln(0+0.05)}{\sqrt{5/6}\times 0.001} = \frac{3.478}{0.00091287} = 3810$$

and with ten Gettysburg readers who all read straight through:

$$z_A = \frac{3.478}{\sqrt{5/15}\times 0.001} = \frac{3.478}{0.00057735} = 6024$$

By contrast, the non-pause features have realistic spreads (median σ in Mode A: rate 0.149 log units, about 15% in speed; loudness 2.6 dB; clarity 4.3 dB; pitch variation 0.28). So real, moderate edits often stay below 3.5:

Take / mode	Planned flaw	Strongest z in the span	Words past 3.5	Result
Gettysburg 02 / A	rushing	+5.26	8 of 26, no run of 3	miss
Reagan 02 / A	rushing	+4.74	3 of 18, no run of 3	miss
Gettysburg 03 / B	monotone	−4.37	1 of 24	miss
Reagan 03 / B	monotone	−3.98	1 of 18	miss
Reagan 03 / A	mumbling	−2.82	0 of 14	miss
Gettysburg 02 / B	too quiet	−2.79	0 of 4	miss
Gettysburg 03 / A	dragging	−2.61	0 of 3	miss

We also checked that no rule was broken: every A/B rule and every Mode C rule fires on fabricated test evidence, and on seven selected development files the target type fires in at least one mode for each type. The misses come from the measured evidence, the thresholds and the three-word rule, not from disabled code. Pause flags do not suppress other types (apart from the three containment rules in Section A.7).

D. The pause timing floor in full

If the aligner's word boundaries are off by about 50 ms, then pause differences smaller than that cannot be trusted. The idea is to never let a boundary's pause spread be smaller than the spread that timing error alone would create. We wrote the protocol and committed it (17a15e4) before writing code, committed the implementation and tests (ae38269) before any run, used exactly one value, ran each split once, and kept a two-hour limit.

D.1 Deriving the floor

Let the true reference pause be p and the timing error ε. Because pauses are compared after T(x) = ln(x + 0.05), an error of ε seconds becomes, to first order (the delta method),

$$T(p+\varepsilon) - T(p) \approx T'(p)\,\varepsilon = \frac{\varepsilon}{p+0.05}$$

so the same 52 ms error matters a lot at a boundary with no pause and very little at a long pause. We set the error to the aligner's measured median boundary error, δ = 0.052 s, and used the exact difference rather than the linear approximation:

$$\sigma_{\text{floor}}(p) = \ln(p+0.052+0.05) - \ln(p+0.05) = \ln\left(1+\frac{0.052}{p+0.05}\right), \qquad \sigma_{\text{eff}} = \max(\sigma,\, \sigma_{\text{floor}})$$

Two choices here are deliberate. First, we use the exact difference because the linear form overstates the floor when ε is not small compared with p + 0.05: at p = 0 it gives 1.04, the exact value is 0.713. Second, 0.052 cannot be used as σ directly, because σ lives in log units and 0.052 is in seconds. The floor applies to pause features only; reference centres, measurements and every other rule stay the same.

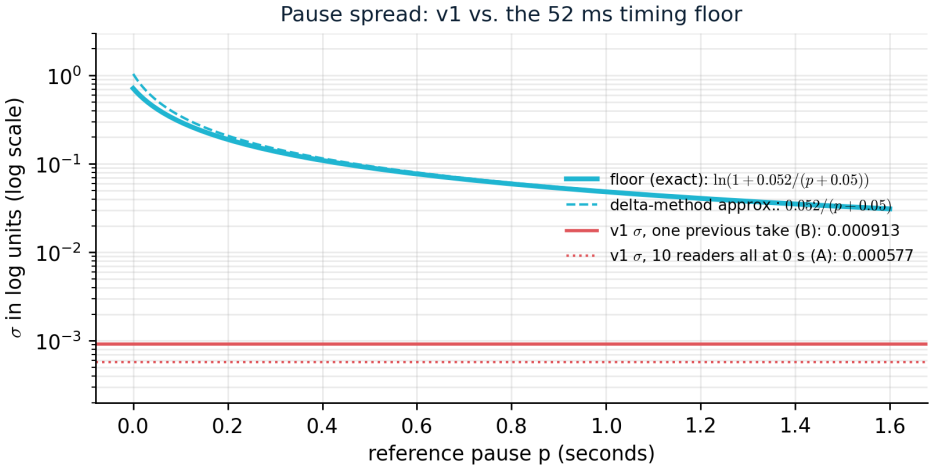


Figure T6. The floor is roughly 780 to 1,240 times larger than the v1 pause spread at a boundary with no reference pause, and shrinks as the reference pause grows.

D.2 What it should do, worked out in advance

A real pause flaw survives. For "larger | sense":

$$z_{\text{floor}} = \frac{3.478}{\ln(0.102/0.05)} = \frac{3.478}{0.713} = 4.88 \ \geq 3.5$$

A 10 ms wobble disappears:

$$0.49 \text{ s vs } 0.48 \text{ s:} \quad z = \frac{\ln(0.54/0.53)}{0.000913} = 20.5 \quad \longrightarrow \quad z_{\text{floor}} = \frac{0.0187}{0.0936} = 0.20$$

Boundaries with real reader spread are untouched. At "here." in Gettysburg take 03 (Mode A), readers pause 1.39 s and σ = 0.257, while the floor is only 0.036, so $\sigma_{\text{eff}} = \sigma$.

Missing-pause flags can never be removed. A missing pause needs reference r ≥ 0.3 s and yours ≤ 0.4r. The smallest possible log difference is ln((r + 0.05)/(0.4r + 0.05)), and dividing it by the floor gives at least 5.21 (at r = 0.3), which is above 3.5 for every r. So when the floor is active the flag stays, and when it is not active nothing changes.

Awkward pauses need to be longer. Where the reference has no pause, the floor-adjusted rule needs ln((p + 0.05)/0.05) ≥ 3.5 × 0.713, that is p ≥ 0.556 s, instead of the 0.45 s length rule. An injected pause of 0.6 s or more at such a boundary still qualifies; where readers already pause, the required length is higher.

D.3 Results (post-hoc; test split previously seen with v1)

Cohort	Version	Precision	Recall [95% CI]	F1 [95% CI]	Clean alarms / min	Hits / misses / FP
Test A-LOSO	v1	0.125	0.400 [0.328, 0.471]	0.190 [0.156, 0.223]	1.341	82 / 123 / 576
Test A-LOSO	v1 + floor	0.139	0.395 [0.324, 0.466]	0.205 [0.168, 0.245]	1.118	81 / 124 / 503
Test B	v1	0.018	0.390 [0.289, 0.494]	0.035 [0.026, 0.045]	0.000	32 / 50 / 1,739
Test B	v1 + floor	0.033	0.366 [0.265, 0.474]	0.061 [0.044, 0.079]	0.000	30 / 52 / 870
Genuine A	v1	0.049	0.308	0.085	8.645*	4 / 9 / 77
Genuine A	v1 + floor	0.056	0.308	0.094	7.893*	4 / 9 / 68
Genuine B	v1	0.048	0.231	0.079	n/a	3 / 10 / 60
Genuine B	v1 + floor	0.067	0.231	0.103	n/a	3 / 10 / 42

* Uses only the two natural take 01 recordings (2.66 minutes). Intervals use the same file-level bootstrap as v1.

- Only **awkward-pause** flags were removed: 74 (A-LOSO), 871 (B), 9 (genuine A), 18 (genuine B). No missing-pause flag was removed, as derived above.
- Rushing, dragging, monotone, too_quiet and mumbling regions are identical in every file. **No non-pause flaw became a hit.**
- Three correct awkward-pause matches were lost on the test set (1 in A-LOSO, 2 in B); none on the genuine takes.
- Precision and F1 rise modestly and false alarms fall, but recall falls slightly and the genuine non-pause misses are untouched.

Decision

The floor is a physically motivated correction that removes many awkward-pause flags caused by tiny timing differences. It is not a validated per-boundary noise model (0.052 s is the observed median aligner error), it was tested after the test set had been seen, it does not touch missing-pause flags, and it does not solve the non-pause misses. So v1 stays the primary result; the floor is worth testing on fresh data in the next frozen version.

E. Next steps, with the mathematics

The limitations in the main report point directly to the work that would turn Speech-Eval from a research prototype into a product people can rely on. The plan has four parts, in order of priority. Each step keeps the rule that made this project trustworthy: decide the method first, freeze it, then test once on data it has never seen.

E.1 Prove that the flags match what people hear

- Collect ratings from at least three independent listeners per recording (the rating pack and scripts already exist, with zero responses so far). Report agreement between raters (Krippendorff's alpha) and between raters and the system.
- Do the human spot-check of the aligner's word boundaries, so the 52 ms error figure rests on listening, not visual marks only.
- Record new speakers, including spontaneous (unscripted) speech, and keep them as a fresh test set that no version has seen.

E.2 Better mathematics

A model for pauses that expects zeros. Most boundaries have no pause, which is why a single log-normal spread collapses to nearly zero (Section C). A two-part model asks two questions: how often do good readers pause here, and how long when they do?

$$\hat{\pi}_j = \frac{k_j + 1}{n + 2} \qquad (\text{share of readers who pause at boundary } j, \text{ smoothed})$$

$$S_j = -\ln P(g_j \mid \text{ref}), \qquad P = 1 - \hat{\pi}_j \ \text{ if } g_j < \tau; \qquad P = \hat{\pi}_j \cdot \text{LogNormal}(g_j;\, \mu_j,\, \sigma_j) \ \text{ if } g_j \geq \tau$$

The surprise score S_j replaces the z-score for pauses. A pause where only 1 of 10 readers paused is surprising but not absurd. Gaps shorter than a threshold τ near the aligner's error (about 0.05 s) count as no pause, so a 10 ms timing wobble scores the same as no pause. This removes the near-zero spread without a separate floor.

Measurement noise in every feature, not only pauses. The reference spread describes how readers differ; the aligner adds its own error on top. Adding the two variances gives every feature an honest denominator; for rate, the delta method gives the alignment part directly:

$$\sigma^2_{\text{total}} = \sigma^2_{\text{readers}} + \sigma^2_{\text{align}}, \qquad \sigma_{\text{align, rate}} \approx \frac{\delta\sqrt{2}}{\text{window duration}} \quad (\text{log units})$$

Control false alarms per recording, not per word. A recording has hundreds of word tests, so a fixed k lets false flags add up. Converting z-scores to p-values and applying the Benjamini–Hochberg procedure fixes the expected share of false flags instead:

$$\text{sort the } M \text{ word p-values; flag the first } m \text{ with } p_{(i)} \le \frac{i}{M}\, q \qquad (q = \text{allowed share of false flags})$$

This targets the failed clean-alarm guard directly. Alternatively, k can be set so that clean development recordings give at most a chosen number of alarms per minute.

- **Learn the prior weight.** n_0 = 5 was fixed by design. It can be estimated from the training readers (empirical Bayes: choose the n_0 that best predicts a left-out reader), and the pooled spread can be computed per word type (sentence end, comma, mid-phrase) instead of one value for all words.
- **Catch gradual changes.** Rushing that builds up over a sentence can miss the three-word rule. A change-point method (for example CUSUM) on the sequence of word z-scores detects a sustained shift instead of isolated peaks.
- **Calibrate severity on listeners.** Replace the fixed severity bins (Spearman 0.157 on development data) with an ordinal regression fitted to listener ratings, so “severity 4” means what people actually hear as worse.
- **Mode B with more history.** Use all of a speaker's earlier takes, not one, so Mode B has a real spread of its own.

E.3 More languages

Most of the pipeline is not tied to English: semitones, decibels, seconds and robust z-scores work in any language. Four parts need care:

- **Alignment.** The Montreal Forced Aligner can use acoustic models and pronunciation dictionaries for other languages; each language needs its own bake-off against hand-marked boundaries, as we did for English.
- **Syllables.** Speaking rate counts syllables from an English dictionary. Each language needs its own syllable counter (from its pronunciation lexicon or grapheme-to-phoneme model).
- **Pitch in tone languages.** In Mandarin or Thai, pitch also carries word meaning, so “monotone” must be measured after removing the expected tone shape of each syllable.
- **Norms and references.** Pausing and rhythm habits differ between languages, so every language needs its own expert readings, its own thresholds chosen on its own development speakers, and native-speaker listeners for validation. Indian languages such as Hindi, Tamil and Telugu would be a natural first step for users in India.

E.4 A better product

- **Any text, any speech.** Mode B for texts the user brings (today it needs a known text), and spontaneous speech with an automatic transcript.
- **Progress over time.** A history view that shows how each flaw type changes across a week of practice.

- **Lighter and private.** A smaller image and a faster aligner, so analysis can run on a laptop or phone without uploading audio, and possibly on free hosting.
- **For teachers and therapists.** A class or caseload view where a teacher or speech-language therapist reviews only the flagged moments.

A possible order of work

Phase	Goal	Done when
1. Validate	Listener ratings, alignment spot-check, fresh test speakers	Agreement measured and reported
2. Fix the maths	Two-part pause model, noise-aware spreads, false-alarm control	Clean-alarm guard met on fresh data
3. Calibrate	Severity and scores fitted to listener ratings	Scores track what people hear
4. Expand	One more language, any-text Mode B, progress view	Same validation repeated per language
5. Pilot	A class and, with therapists, a clinic	Feedback from real users

Each phase is frozen and tested before the next begins, so every claim stays as checkable as the ones in this report.

F. Every fixed constant

Constant	Value	Where it is used
Pitch / intensity grid	10 ms	Praat feature extraction
Clarity bands	2–6 kHz over 0–2 kHz	high-band energy ratio
Window	word ± 1 neighbour, within sentence	rate, pitchvar, loud, clarity
Pause excluded from rate	> 0.25 s	rate denominator
Transform offsets	pause +0.05 s; pitchvar +0.25 st; Mode C range +0.5 st	log transforms
MAD scale	1.4826	robust spread
Prior weight n_0	5	shrinkage
Numerical safeguard	0.001 (transformed units)	pooled spread
Threshold k (A/B)	3.5 for all seven types	word flags
Threshold k (Mode C)	2.0	sentence flags
Region rule	≥ 3 flagged words, ≤ 1-word gap, one sentence	non-pause regions
Awkward pause	$z \geq 3.5$ and ≥ 0.45 s	pause events
Missing pause	$z \leq -3.5$, reference ≥ 0.3 s, yours ≤ 40%	pause events
Short pause event	widened to 30 ms	display and matching
Severity cut points	2.0, 2.75, 3.5, 4.059, 6591	severity 1–5
Rubric	duration ×10; event 0.4 per severity; weights 0.25/0.25/0.20/0.15/0.15	score card
Matching	IoU ≥ 0.3; pause centres ≤ 0.25 s (synthetic)	evaluation
Bootstrap	1,000 resamples, seed 20261001, percentile	confidence intervals
Pause floor (post-hoc only)	$\delta = 0.052$ s	Section D

G. Notes on consistency

- The diagnosis tables in the repository label pitch variation as “log +0.5”. The A/B code uses ln(x + 0.25) for pitch variation; +0.5 is the Mode C pitch-range offset. All numbers in this report come from the code.
- An early acceptance note expected a single-reference spread within 1% of the pooled spread. The code gives $\sqrt{(5/6)} \approx 0.913$ of it, about 8.7% lower. All published results use the code.
- Because flagged words have $|z| \geq 3.5$, severities 1 and 2 cannot occur in Modes A and B; they can occur in Mode C ($k = 2.0$).